

QINGLONG, China

WMO: 544360

Lat: 40.4087N

Lon: 118.9522E

Elev: 255

StdP: 98.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-15.2	-28.5	0.3	-8.9	-26.3	0.4	-8.4	6.8	-4.0	5.8	-3.6	0.6	220	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	32.9	21.8	31.5	21.6	30.3	21.4	25.8	29.4	24.9	28.3	24.1	27.3	3.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.4	28.0	23.9	19.4	27.0	23.1	18.4	26.4	81.6	29.5	77.7	28.5	74.3	27.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(4)
(5) 6.7	5.6	4.9	DB	-20.1	35.5	2.5	1.7	-21.9	36.7	-23.4	37.7	-24.8	38.6	-26.7	39.8	(4)
			WB	-20.6	26.9	2.4	0.9	-22.3	27.5	-23.7	28.0	-25.0	28.4	-26.7	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-6.7	-2.9	4.2	12.1	18.2	22.4	25.0	23.9	18.6	10.9	2.2	-4.7	(6)	
	DBStd	11.62	3.55	3.94	4.20	4.05	3.26	2.53	2.09	2.35	3.15	3.84	4.23	3.58	(7)	
	HDD10.0	1813	519	360	186	24	1	0	0	0	0	35	233	456	(8)	
	HDD18.3	3506	777	593	438	189	42	2	0	1	34	232	483	714	(9)	
	CDD10.0	1939	0	0	6	87	254	373	466	431	259	63	1	0	(10)	
	CDD18.3	591	0	0	0	2	38	125	208	173	43	1	0	0	(11)	
	CDH23.3	5401	0	0	1	45	512	1197	1849	1412	373	12	0	0	(12)	
(13) Wind	CDH26.7	1773	0	0	0	4	138	442	664	448	74	1	0	0	(13)	
	WSAvg	1.8	1.4	1.8	2.3	2.5	2.3	1.9	1.6	1.5	1.5	1.6	1.6	1.4	(14)	
	PrecAvg	715	2	4	9	26	43	96	232	196	64	32	9	4	(15)	
	PrecMax	1129	14	34	27	108	144	233	605	532	231	128	27	24	(16)	
	PrecMin	475	0	0	0	0	9	15	87	35	0	0	0	0	(17)	
	PrecStd	196	4	6	8	25	30	58	116	119	44	32	7	6	(18)	
	DB	5.9	11.3	20.6	27.0	32.3	34.8	35.3	33.4	30.6	25.0	16.5	7.4		(19)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	-0.1	3.4	10.1	13.7	17.7	19.1	22.8	22.9	20.0	16.0	9.0	1.6		(20)	
	DB	3.3	8.7	17.3	24.5	30.0	32.5	33.1	31.6	28.7	22.1	14.1	5.1		(21)	
	MCWB	-1.8	2.0	8.0	13.0	16.4	19.8	23.2	23.0	18.9	14.1	7.5	0.6		(22)	
	DB	1.6	6.7	14.6	22.5	28.1	30.9	31.6	30.4	27.1	20.2	11.8	3.3		(23)	
	MCWB	-2.8	0.9	6.3	11.8	15.8	19.4	23.1	22.8	18.4	12.7	5.9	-0.7		(24)	
	DB	-0.1	4.8	12.2	20.4	26.3	29.3	30.2	29.2	25.5	18.3	9.6	1.7		(25)	
	MCWB	-3.8	-0.2	4.8	10.7	15.1	19.2	23.0	22.4	17.8	11.6	4.8	-1.8		(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	0.2	4.7	11.3	16.6	20.4	23.7	27.0	26.7	23.3	17.6	10.5	3.5		(27)	
	MCDB	4.8	8.7	18.7	23.4	27.9	28.8	30.6	30.2	27.3	22.0	14.1	6.0		(28)	
	WB	-1.3	2.8	8.8	14.8	18.8	22.5	26.1	25.6	21.6	16.1	8.4	1.2		(29)	
	MCDB	2.4	7.4	15.6	21.5	26.5	27.8	29.9	29.1	25.7	20.4	12.4	4.0		(30)	
	WB	-2.5	1.3	7.0	13.4	17.7	21.6	25.3	24.7	20.3	14.5	6.9	-0.3		(31)	
	MCDB	1.2	6.4	13.8	20.3	24.9	26.8	28.9	28.0	24.1	18.2	10.8	2.8		(32)	
	WB	-3.7	-0.2	5.2	11.8	16.6	20.8	24.5	23.9	19.2	13.0	5.2	-1.7		(33)	
(34) Mean Daily Temperature Range	MCDB	-0.3	4.6	11.4	18.4	23.3	26.0	28.0	26.9	23.4	16.6	9.3	1.6		(34)	
	MDBR	12.1	12.0	11.9	12.3	13.0	11.3	8.9	9.6	12.1	12.7	11.2	11.4		(35)	
	5% DB	13.8	14.6	15.2	15.6	16.6	15.2	11.9	11.3	14.1	15.1	14.1	13.0		(36)	
	MCWBR	9.6	9.2	8.4	7.0	6.2	5.1	4.2	4.2	5.5	7.4	8.3	9.0		(37)	
	5% WB	13.1	13.5	14.4	13.1	13.2	10.6	8.0	8.1	10.2	12.0	12.8	12.0		(38)	
	MCWBR	9.2	8.8	8.3	6.8	6.0	4.5	3.7	3.7	4.9	6.9	8.4	8.6		(39)	
	taub	0.383	0.466	0.508	0.597	0.640	0.759	0.754	0.674	0.606	0.575	0.448	0.394		(40)	
(41) Clear-Sky Solar Irradiance	taud	2.040	1.857	1.805	1.675	1.621	1.474	1.520	1.646	1.718	1.721	1.939	2.042		(41)	
	Ebn at Noon	729	710	732	699	681	604	600	635	642	595	643	677		(42)	
	Edh at Noon	119	166	195	237	256	297	281	241	211	188	131	110		(43)	
	RadAvg	2.59	3.45	4.70	5.42	6.08	5.43	4.95	4.93	4.45	3.44	2.57	2.27		(44)	
(45) All-Sky Solar Radiation	RadStd	0.17	0.26	0.32	0.30	0.39	0.47	0.43	0.32	0.28	0.26	0.28	0.15		(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+68	N/A

Nomenclature: See separate page